

HR DATABASE MANAGEMENT SYSTEM

Executive Summary

This project develops an HR database analysis using SQL Server concepts and a relational HR sample database. The supplied database covers employee, job, department, dependent, location, country and region information, with SQL tasks spanning retrieval, filtering, joins, aggregation, set operators, CASE expressions, data modification and advanced subqueries.

- Listed annual payroll = 314,100; average salary = 8,053.85; median salary = 7,700.
- Shipping has the largest headcount at 7; Sales, Purchasing and Finance each have 6.
- Executive has the highest departmental average salary at 19,333.33.
- Executive and Sales together account for 36.8% of listed payroll.
- 27 of 39 employees fall in the SQL-defined High-salary category (>5,000).
- 6 employees have no phone number.
- All 39 supplied employee salaries fall within the defined salary range of their assigned job.

Management Interpretation

The analytical layer converts operational HR records into management information: headcount shows workforce concentration, salary analysis shows payroll distribution, manager-span analysis shows reporting structure, and location joins establish organizational footprint. Because this is a small sample database, the results are analytical demonstrations rather than current workforce decisions.

Dataset & Database Architecture

Table	Purpose	Rows stated in brief
employees	Employee master data	40
dependents	Employee dependent records	30
departments	Department master data	11
jobs	Job titles and salary bands	11
locations	Department locations	7
countries	Country master data	25
regions	Regional grouping	4

The project brief describes seven core tables: employees, jobs, departments, dependents, locations, countries and regions. The SQL solution creates these tables with primary and foreign-key relationships. Employees link to jobs and departments and self-reference through manager_id; dependents link back to employees; the location hierarchy links departments to locations, countries and regions. The supplied SQL contains 19 job records although the brief states 11. This should be reconciled before describing the database as a finalized production dataset.

Table 1: Dataset

Relationship Map

- regions → countries → locations → departments → employees
- jobs → employees
- employees → employees (manager_id reporting hierarchy)

- employees → dependents

Analytical Methodology

Analytical layer	SQL techniques	Purpose
Data retrieval	SELECT, aliases, TOP, DISTINCT	Retrieve and inspect HR records
Filtering	WHERE, ORDER BY, LIKE, IN, BETWEEN, NULL	Segment employees and identify exceptions
Logical operators	AND, OR, NOT, ALL, IS NULL	Apply multi-condition HR rules
Joins	INNER, LEFT, SELF, FULL OUTER, CROSS	Combine workforce and organizational data
Aggregation	GROUP BY, COUNT, SUM, MIN, MAX, AVG	Build headcount and compensation metrics
Set/subquery logic	UNION, UNION ALL, INTERSECT, EXISTS, subqueries	Compare populations and answer nested questions
Classification	CASE	Create salary and tenure categories
Maintenance	UPDATE, ALTER TABLE, foreign keys	Demonstrate controlled database maintenance

The completed SQL combines HR-specific queries with instructional demonstrations. The report focuses on findings that can be interpreted from the HR tables while identifying illustrative demo-table examples separately.

Table 2: HR KPI Dashboard

HR KPI Dashboard

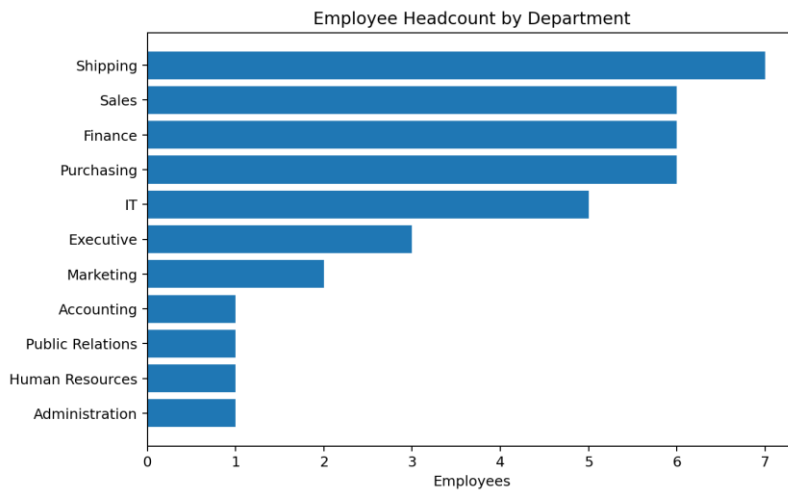
39	11	314,100	8,053.85	7,700	6
Employees in supplied SQL	Departments	Listed payroll	Average salary	Median salary	Missing phone numbers

Table 3: HR KPI Dashboard

Workforce Structure Analysis

ID	Department	Headcount	Avg Salary	Total Salary	Payroll Share
5	Shipping	7	5,885.71	41,200	13.1%
8	Sales	6	9,616.67	57,700	18.4%
3	Purchasing	6	4,150.00	24,900	7.9%
10	Finance	6	8,600.00	51,600	16.4%
6	IT	5	5,760.00	28,800	9.2%
9	Executive	3	19,333.33	58,000	18.5%
2	Marketing	2	9,500.00	19,000	6.0%
1	Administration	1	4,400.00	4,400	1.4%
4	Human Resources	1	6,500.00	6,500	2.1%
7	Public Relations	1	10,000.00	10,000	3.2%
11	Accounting	1	12,000.00	12,000	3.8%

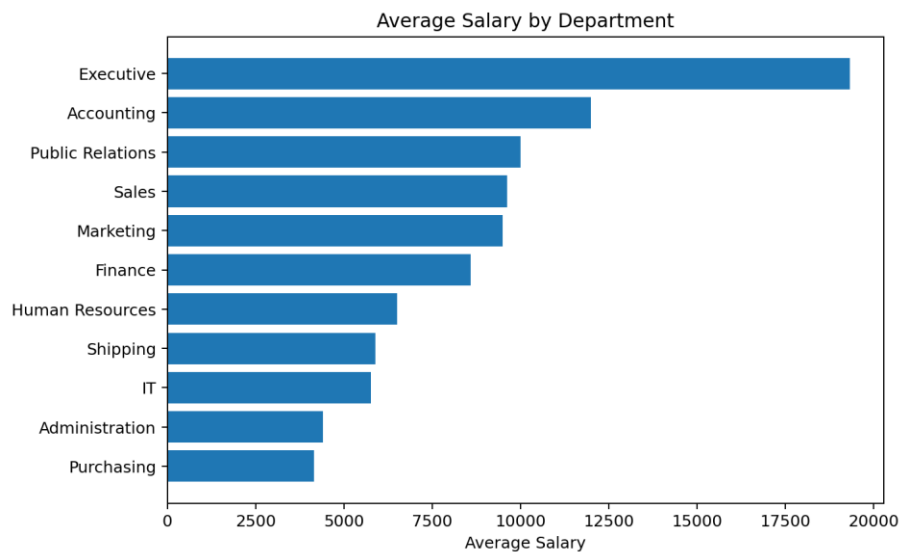
Table 4: Workforce Structure



Shipping has the highest listed headcount (7), followed by Sales, Purchasing and Finance with 6 each. The four largest departments contain 25 of 39 listed employees (64.1%).

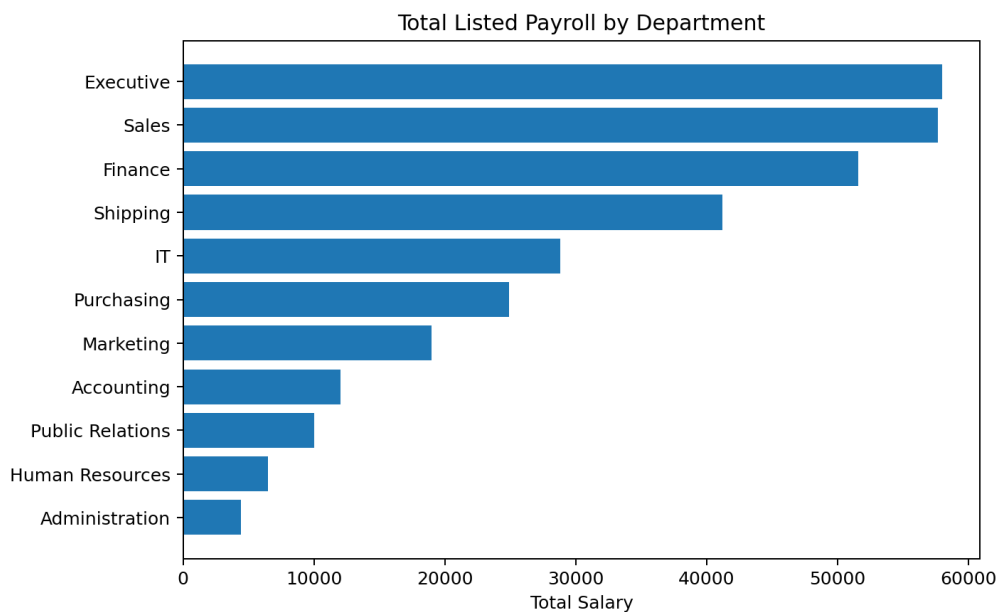
Chart 1: Workforce Structure

Compensation Analysis



Overall average salary is 8,053.85, compared with a median of 7,700. The highest listed salary is 24,000. Executive has the highest departmental average at 19,333.33.

Chart 2: Average Salary by department



Executive, Sales and Finance together contribute 53.3% of listed payroll, reflecting both compensation levels and department size.

Chart 3: Payroll by department

Salary Classification

Category	Rule	Employees	Share
Low	< 3,000	5	12.8%
Average	3,000–5,000	7	17.9%
High	> 5,000	27	69.2%

Table 5: Salary Classification

All 39 employee salaries fall between the minimum and maximum salary defined for their assigned job. This is a useful master-data validation check, although it does not establish pay equity or market competitiveness.

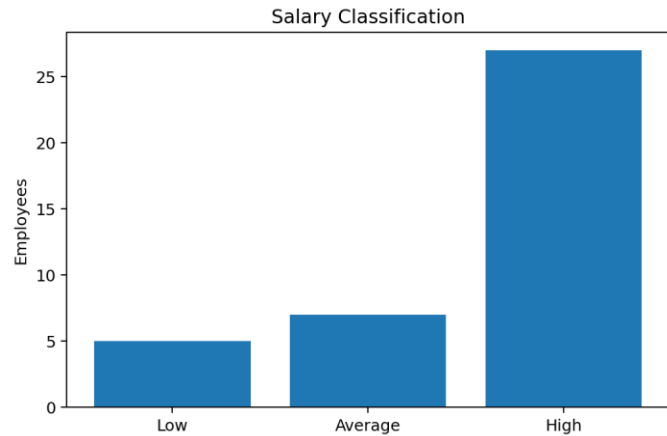


Chart 4: Salary Classification

Management & Reporting Structure

Manager ID	Manager Name	Direct Reports
100	Steven King	14
101	Neena Kochhar	5
108	Nancy Greenberg	5
114	Den Raphaely	5
103	Alexander Hunold	4
123	Shanta Vollman	2
102	Lex De Haan	1
120	Matthew Weiss	1
201	Michael Hartstein	1

Table 6: Management



Chart 5: Management

The self-join identifies manager-report relationships. Manager ID 100 has 14 direct reports, while three other managers have 5 each. These are descriptive structural observations, not measures of managerial effectiveness.

Employee Dependents & Data Quality

Check	Result	Interpretation
Employee rows in SQL	39	39 employee INSERT rows
Employee rows stated in brief	40	One-record discrepancy
Dependent rows	30	Matches brief count
Orphan dependent employee ID	206	Referenced but absent from supplied employee rows
Employees with missing phone	6	Contact field incomplete for some records
Salary outside job range	0	No out-of-band salary in supplied employee rows

Table 7: Interpretation

The dependent table contains 30 rows. 29 distinct employee IDs in that table match the 39 supplied employee records; employee ID(s) 206 do not. Because the schema defines a foreign key from dependents.employee_id to employees.employee_id, the orphan reference to employee 206 is a material integrity issue if the supplied script is executed exactly as written. The missing record should

be reconciled against the authoritative source rather than guessed.

SQL Task Coverage & Analytical Interpretation

Module	Techniques	Business interpretation
Task 1	SELECT, ORDER BY, DISTINCT, TOP, WHERE	Core retrieval, ranking and filtering
Task 2	Logical/special operators	Multi-condition segmentation, NULL handling, pattern matching
Task 3	INNER, LEFT, SELF, FULL OUTER, CROSS JOIN	Relational integration and hierarchy analysis
Task 4	GROUP BY, HAVING	Headcount, salary and manager-span metrics
Task 5	UNION, UNION ALL, EXISTS, CASE, UPDATE	Set logic, existence checks, classification and maintenance
Final Task	Subqueries, ALL, EXISTS, derived tables	Nested location, salary and departmental analysis

Highest listed salary = 24,000. 16 employees earn above the overall average. Only 1 employee earns above the highest departmental average (19,333.33). The > ALL departmental-minimum query produces a threshold based on the highest departmental minimum salary (17,000).

Table 8: Analytical Interpretation

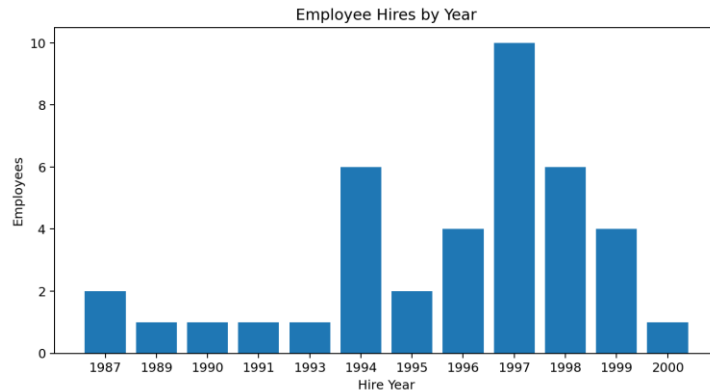
Geographic & Organizational Footprint

Department	City	Country	Region
Administration	Seattle	United States of America	Americas
Marketing	Toronto	Canada	Americas
Purchasing	Seattle	United States of America	Americas
Human Resources	London	United Kingdom	Europe
Shipping	South San Francisco	United States of America	Americas
IT	Southlake	United States of America	Americas
Public Relations	Munich	Germany	Europe
Sales	Oxford	United Kingdom	Europe
Executive	Seattle	United States of America	Americas
Finance	Seattle	United States of America	Americas
Accounting	Seattle	United States of America	Americas

Table 9: Geographic & Organizational Footprint

The sample connects departments to locations in the United States, Canada, the United Kingdom and Germany, demonstrating how HR reporting can combine organizational and geographic master data.

Hiring Pattern Analysis



Hire dates span 1987-06-17 to 2000-01-04. The largest hiring year in the supplied sample is 1997, with 10 listed hires. This is a historical sample pattern, not a current recruitment trend.

Chart 6: Hiring Pattern Analysis

Limitations & Assumptions

- This is a small sample HR database, not evidence of a live company's current workforce.
- The supplied brief and SQL disagree on employee count (40 stated vs 39 supplied) and job count (11 stated vs 19 supplied).
- Employee 206 is referenced by a dependent but is absent from the supplied employee INSERT records.
- Salary analysis excludes bonuses, benefits, overtime, currency conversion and other total-compensation components.
- No performance, attrition, attendance, recruitment or engagement fields are supplied.
- The SQL UPDATE changes Sarah Bell's surname to Lopez and then synchronizes dependent surnames; this is treated as a demonstrated database-maintenance operation.
- The FULL OUTER JOIN and CROSS JOIN examples use separate demonstration tables and are not core HR master data.

Conclusion

The project demonstrates an end-to-end SQL-based HR analytics workflow: relational schema creation, workforce retrieval and filtering, organizational joins, headcount and compensation aggregation, manager hierarchy analysis, salary classification and advanced subqueries. The main analytical themes are workforce distribution, payroll concentration, reporting structure, geographic footprint and data quality.